

Use Case Flow

Smart Expense & Budget Management System

Name: _____ SRN: _____ Section: _____

Use Case ID	UC-03
Use Case Name	Manage Expense Records (adding an expense)
Primary Actor	User
Other Actors	Analytics / Prediction Engine (category suggestion and anomaly check), Notification Service (budget alert)
Relationships	<<include>> Validate & Categorize Entry; <<extend>> Suggest Expense Category, Detect Unusual Spending
Trigger	User taps “Add Expense” on the dashboard
Requirements Covered	FR03, FR04, FR05, FR08, FR09, FR11, SR04, SR05

Preconditions

1. User has an account and is logged in with a valid session.
2. At least one expense category exists, either default or created by the user.
3. The app can reach the server.

Postconditions

1. The expense is saved against the logged-in user with amount, date and category.
2. Remaining budget for the month and for that category is recalculated.
3. Charts and the dashboard reflect the new entry.
4. If the use case fails, no partial expense record is left behind.

Main Success Scenario

1. User taps “Add Expense”.
2. System shows the expense form with today’s date filled in.
3. User enters the amount and a short note or merchant name.
4. System asks the Analytics Engine for a category suggestion and shows it (<<extend>> Suggest Expense Category). The user can keep it or pick a different one.
5. User confirms the category and submits.
6. System runs <<include>> Validate & Categorize Entry, checking that the amount is a positive number, the date is not in the future and a category is set.
7. Validation passes and the expense is saved against the user’s account.
8. System recalculates the monthly and category-wise remaining budget.
9. System sends the entry to the Analytics Engine for an anomaly check (<<extend>> Detect Unusual Spending).
10. System shows the updated remaining budget and the new entry at the top of the list.
11. Use case ends successfully.

Alternate Flow 6a: Validation Fails

Happens at step 6 if the amount, date or category is not acceptable.

- 6a1. System does not save the expense and marks the fields that are wrong, for example “Amount must be greater than zero”.
- 6a2. User corrects them and submits again. Flow goes back to step 6.
- 6a3. If the user leaves the screen instead, nothing is saved.

Alternate Flow 8a: Budget Is Crossed

Happens at step 8 if this expense takes the user past the monthly or category budget.

- 8a1. System saves the expense as normal, since the entry itself is valid.
- 8a2. System shows a budget alert on screen and asks the Notification Service to send one (FR09).
- 8a3. Remaining budget is shown as a negative figure so the user can see how far over they are.
- 8a4. Flow continues from step 9.

Alternate Flow 9a: Expense Looks Unusual

Happens at step 9 if the amount is well outside the user’s normal range for that category.

- 9a1. System flags the expense for review and shows a note asking the user to confirm the amount.
- 9a2. If the user confirms, the flag is cleared and the expense stays as it is.
- 9a3. If the user says it is a mistake, the form reopens for editing and the flow goes back to step 3.